Deep Underwater Image Quality Assessment With Explicit Degradation Awareness Embedding

Qiuping Jiang, *Senior Member, IEEE*, Yuese Gu, Zongwei Wu, *Member, IEEE*, Chongyi Li, *Senior Member, IEEE*, Huan Xiong, Feng Shao, *Member, IEEE*, and Zhihua Wang, *Member, IEEE*

***Abstract*—Underwater Image Quality Assessment (UIQA) is currently an area of intensive research interest. Existing deep learning-based UIQA models always learn a deep neural network to directly map the input degraded underwater image into a final quality score via end-to-end training. However, a wide variety of image contents or distortion types may correspond to the same quality score, making it challenging to train such a deep model merely with a single subjective quality score as supervision. An intuitive idea to solve this problem is to exploit more detailed degradation-aware information as supplementary guidance to facilitate model learning. In this paper, we devise a novel deep UIQA model with Explicit Degradation Awareness embedding, i.e., EDANet. To train the EDANet, a two-stage training strategy is adopted. First, a tailored Degradation Information Discovery subnetwork (DIDNet) is pre-trained to infer a residual map between the input degraded underwater image and its pseudoreference counterpart. The inferred residual map explicitly characterizes the local degradation of the input underwater image. The intermediate feature representations on the decoder side of DIDNet are then embedded into the Degradation-guided Quality Evaluation subnetwork (DQENet), which significantly enhances the feature characterization capability with higher degradation awareness for quality prediction. The superiority of our EDANet against 18 state-of-the-art methods has been well demonstrated by extensive comparisons on two benchmark datasets. The source code of our EDANet is available at https://github.com/yia-yuese/EDANet.**

***Index Terms*—Image quality assessment, underwater image, degradation awareness, deep learning.**

Received 22 May 2023; revised 5 April 2024; accepted 27 January 2025. Date of publication 20 February 2025; date of current version 25 February 2025. This work was supported in part by the Natural Science Foundation of Zhejiang under Grant LR22F020002; in part by the Natural Science Foundation of China under Grant 62271277 and Grant 62471263; in part by the Natural Science Foundation of Ningbo under Grant 2022J081 and Grant 2022J067; in part by the Key Research and Development Program of Ningbo under Grant 2024Z292 and Grant 2024Z113; and in part by the "Leading Goose" Research and Development Program of Zhejiang under Grant 2025C01069. The associate editor coordinating the review of this article and approving it for publication was Prof. Joao M. Ascenso. *(Corresponding authors: Zongwei Wu; Zhihua Wang.)*

Qiuping Jiang, Yuese Gu, and Feng Shao are with the Faculty of Information Science and Engineering, Ningbo University, Ningbo 315211, China (e-mail: jiangqiuping@nbu.edu.cn).

Zongwei Wu is with the Computer Vision Laboratory, University of Würzburg, 97074 Wurzburg, Germany (e-mail: zongwei.wu.97@gmail.com).

Chongyi Li is with the School of Computer Science, Nankai University, Tianjin 300071, China (e-mail: lichongyi25@gmail.com).

Huan Xiong is with the Institute for Advanced Study in Mathematics, Harbin Institute of Technology, Harbin 150006, China (e-mail: huan.xiong.math@gmail.com).

Zhihua Wang is with the Department of Computer Science, City University of Hong Kong, Hong Kong (e-mail: zhihua.wang@my.cityu.edu.hk).

Digital Object Identifier 10.1109/TIP.2025.3539477

I. Introduction

UNDERWATER imaging is commonly employed to record the visual information of the marine environment, playing a vital role in various marine-related applications such as marine ecological protection, marine resource development, and marine geological exploration [1], [2]. However, the complicated underwater imaging environment leads to the degradation of captured images due to wavelength-dependent absorption and scattering [3], [4], [5], [6], [7]. This degradation also leads to various kinds of quality issues, such as color cast, reduced contrast, poor visibility, etc. Therefore, it is essential to develop an accurate objective image quality assessment (IQA) algorithm capable of efficiently predicting underwater image quality, not only for online quality monitoring during the imaging stage, but also to guide the quality enhancement process [8], [9] after the images are captured.

In the literature, objective IQA algorithms can be roughly divided into two categories: reference-based [10], [11], [12], [13], [14], [15] and non-reference-based (NR) [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31]. Although reference-based IQA methods usually demonstrate superior performance, accessing reference images is often unavailable in many practical scenarios. This issue is particularly pertinent for real-world underwater scenes, where only distorted images are provided, and thus only the NR paradigm can be applied.

As is well known, NR-IQA is a challenging problem that has received considerable attention over the past decade. Traditional NR-IQA methods [16], [17], [18], [19], [20], [21], [22], [23], [24] focus on the extraction of handcraft features and rely on prior knowledge by modeling the properties of the human visual system [32] or the regularities of Natural Scene Statistics (NSS). However, the efficacy of the traditional NR-IQA methods is limited to synthetic distortions. Recently, deep neural networks (DNN) have shown significant potential in various vision tasks, including NR-IQA [25], [26], [27], [28], [29], [30], [31]. Compared to traditional methods [16], [17], [18], [19], [20], [21], [22], [23], [24], deep NR-IQA methods can jointly optimize feature learning and quality regression in an end-to-end manner, yielding more accurate prediction results. However, a significant challenge for NR-IQA is that images with different contents or different types of distortion can be assigned with the same quality score, as shown in Fig. 1. Therefore, it is challenging and it remains unclear how to map the intermediate feature representations with large variations to the same target quality score. Existing deep

1941-0042 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.

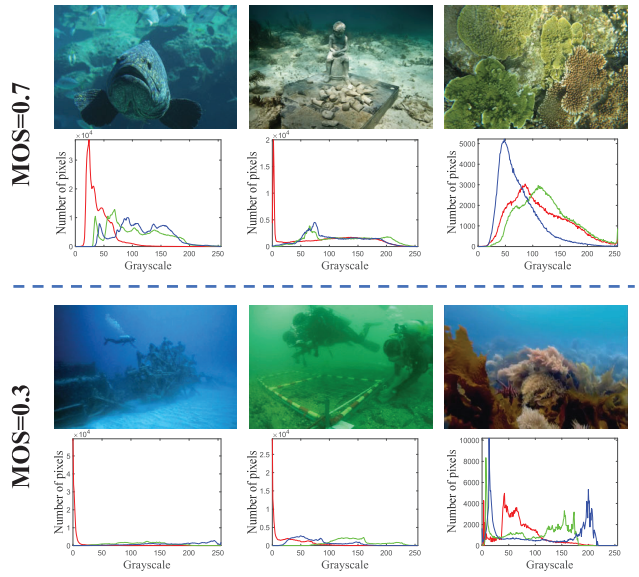


Fig. 1. **Observation and motivation**. Underwater images with different image contents and degradation types can be assigned with the same quality score, which is a big challenge for learning the relationship between feature representations and quality scores.

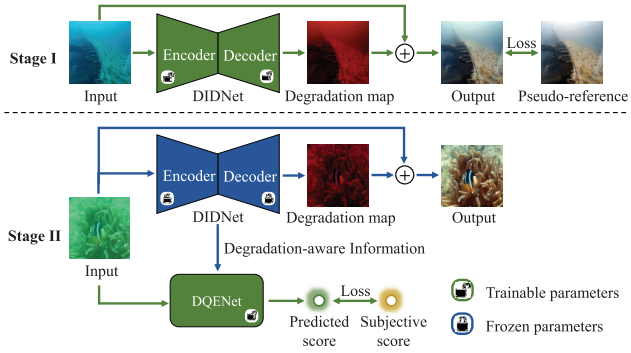


Fig. 2. **Training strategy**. The training of our proposed EDANet involves two stages. Initially, we train a DIDNet for residual learning on the existing underwater image enhancement (UIE) datasets. The goal of the DIDNet is to estimate the residual maps between the input degraded underwater images and their corresponding pseudo-reference images. Then, the degradation-aware information acquired by the DIDNet is embedded to guide the training of DQENet. In this figure, the modules in **green** are trainable, i.e., the DIDNet in stage I and the DQENet in stage II. In contrast, the module in **blue** is frozen, i.e., the DIDNet in stage II.

NR-IQA models train the network merely using the subjective quality score as supervision. However, this kind of supervision is somehow insufficient, especially for the UIQA problem, where only relatively small datasets are available for network training.

To address the above challenge, we made an attempt to exploit additional degradation-aware information as supplementary guidance to support learning a more effective deep UIQA model. We observe that as light undergoes absorption and refraction in the water media, raw underwater images usually suffer from color casts, reduced contrast, poor visibility, etc. Each of these quality defects will degrade the overall quality, and a local degradation map characterizing the distribution and degree of all these quality defects can provide more detailed degradation-aware information. Therefore, in this work, we propose a novel Explicit Degradation Awareness embedding-based deep UIQA model. The proposed model is called EDANet in short. Specifically, given an input underwater image, EDANet first exploits explicit degradation-aware information from it by estimating a corresponding local degradation map via a Degradation Information Discovery sub-network (DIDNet) and then embeds the intermediate representations obtained from the DIDNet into the subsequent Degradation-guided Quality Evaluation sub-network (DQENet) to facilitate quality prediction.

In order to obtain explicit degradation-aware information for an input underwater image, the oracle scenario would have to access the non-distorted images as reference. However, non-distorted underwater images are usually not available in real-world underwater scenarios. Previous studies in multitask learning have demonstrated that knowledge learned in one task can be beneficial for other closely related tasks [33], [34], [35]. Therefore, in this work, we resort to the underwater image enhancement (UIE) task to facilitate the UIQA task. As illustrated in Fig. 2, the training of our proposed EDANet involves two stages. Initially, we train a DIDNet on the existing UIE dataset to estimate a residual map between the input degraded underwater image and its corresponding pseudoreference counterpart. The inferred residual map explicitly characterizes the local degradation of the input underwater image. Then, the intermediate feature representations in the decoder side of the DIDNet are embedded into the Degradation-guided Quality Evaluation subnetwork (DQENet), which significantly enhances the feature characterization capability with higher degradation awareness for quality prediction.

Our work differs from conventional deep NR-IQA [25], [26], [27], [28], [29], [30], [31] and domain adaptation approaches [36], [37], [38] in several key aspects. Firstly, unlike conventional methods that rely solely on end-to-end optimization and subjective quality scores, EDANet enriches the learning procedure with intricate insights by leveraging degradation-aware information from the closely-associated UIE task. This approach results in more precise quality predictions and enhanced model training. By furnishing more informative degradation-aware intermediate representations, EDANet simplifies quality prediction and distinguishes itself by effectively tackling intricate image variability and quality defects. Secondly, since the UIE task [39] is closely related to the UIQA task, the domain gap is smaller compared to the traditional domain adaptation problems [40], [41]. Therefore, it is possible to directly transfer the knowledge in an off-the-shelf manner, without adding more regularization modules for domain adaption. Finally, we embed the acquired degradation-aware intermediate representations into our DQENet to guide the end-to-end learning process. Unlike previous works, our EDANet is the first of its kind to output both a degradation map and a single quality score. In such a manner, we can provide quantifiable means to support and better understand visual quality. To conclude, our work makes the following contributions.

- We propose a novel Explicit Degradation Awareness embedding-based deep UIQA model called EDANet which generates an informative degradation-aware intermediate representation to guide the quality prediction

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.

process. This framework has not previously been explored in the UIQA domain, to the best of our knowledge.
- We exploit the degradation-aware information by pretraining a DIDNet for residual learning on the existing UIE datasets and take the intermediate representations in the decoder side of the DIDNet as supplementary information for quality prediction.
- Our EDANet is the first of its kind in the field of UIQA to output a detailed degradation map identifying the local distorted regions as well as a single quality score indicating the overall distortion level.
- The effectiveness of our proposed EDANet has been validated through extensive experiments on two publicly available UIQA databases. Our EDANet, guided by the explicit degradation awareness, outperforms the state-of-the-art NR-IQA and UIQA methods.

The rest of this paper is arranged as follows. The related works are described in Section II. The details of the proposed EDANet are illustrated in Section III. The experimental results are presented and analyzed in Sections IV and V. Finally, conclusions are drawn in Section VI.

II. Related Works

In this section, we will briefly review existing UIQA methods and deep learning-based NR-IQA methods, especially those related to our work.

A. UIQA Methods

Due to the difficulty in obtaining reference underwater images, UIQA metrics always work in NR settings [42], [43]. Nonetheless, due to the dramatically different imaging principles, directly applying the early NR-IQA methods to underwater scenarios is non-trivial. Thus, a number of methods are carefully and specially designed by considering the underwater characteristics. Specifically, Yang and Sowmya [44] proposed an underwater color image quality evaluation method (UCIQE) by calculating weighted summation of the standard deviation of chroma, the contrast of luminance, and the average of saturation. Inspired by human visual system, Panetta et al. [45] proposed an underwater image quality method (UIQM) by combining weighted summation of colorfulness, sharpness, and contrast measures. Similarly, Wang et al. [46] introduced another method based on the measurements of image colorfulness, contrast, and fog density (CCF). On considering the underwater image degradation effects and human visual perception property, FDUM [47] was designed by combining the colorfulness, contrast, and sharpness clues. After analyzing the main factors affecting the visual quality of underwater images, NUIQ [48] was proposed by extracting quality-aware features from the chrominance and luminance components and training a regression model to evaluate the underwater image quality. Zheng et al. [49] proposed an Underwater Image Fidelity (UIF) metric for objective evaluation of enhanced underwater images by exploiting the statistical features in the CIELab space in terms of naturalness, sharpness, and structure. Yang et al. [50] developed a novel ternary classification transformer to predict a quality comparison of underwater images without reference. Wang et al. [51] introduced a new generation-based joint luminance-chrominance underwater image quality evaluation (GLCQE) network. Li et al. [52] devised a novel no-reference UIQA method called MSAEQA based on multiscale and antagonistic energy distribution. Liu et al. [53] proposed a novel quality assessment model for underwater images by in-depth analysis and characterization of image luminance, color cast, sharpness, contrast, fog density and noise to comprehensively describe the underwater image quality.

In common settings, existing UIQA methods can provide reasonable results. However, we observe that when dealing with challenging cases, such as with severe color cast and/or significantly reduced visibility, these methods fail to perform accurately, leaving a large space for further improvement.

B. Deep NR-IQA Methods

Recently, DNNs have shown their exceptional capability of automatically learning feature representation from data, which has also been proven to be powerful to address the NR-IQA problem. For example, Kang et al. [25] developed a pioneering convolution neural network (CNN) for NR-IQA, and achieved a performance leap over traditional feature engineering-based NR-IQA methods. Later, Hou et al. [54] employed a deep belief network (DBN) to classify images into five quality grades and designed a quality pooling strategy to obtain quality scores. Their motivation came from the observation that humans prefer to conduct evaluations qualitatively rather than numerically. To solve the problem of patchwise training, Bosse et al. [26] proposed an end-to-end CNN-based NR-IQA model by adopting a local weighting module. Liu et al. [55] proposed an NR-IQA method taking advantage of ranking loss, since it is often easier and more reliable to sort two images by comparing their quality rather than directly assigning the image to an absolute quality score. To better exploit hierarchical features, Wu et al. [27] introduced a cascaded CNN for NR-IQA. The model was trained on a large-scale dataset that contains more than one million distorted image samples along with their corresponding pseudo-quality labels (obtained by standard FR-IQA methods). To handle both authentic and synthetic distortion types, Zhang et al. [28] proposed a deep bilinear CNN-based NR-IQA model that consists of two CNNs, each of which specializes in one distortion scenario, that is, the authentic distortion and the synthetic distortion. TReS [30] is a deep NR-IQA method that combines CNN and transformer architectures, extracting both local and non-local features from image quality prediction. In TTDA [31], considering that the distribution shift between the test scenario and the training scenario often leads to poor performance of the model trained in the training scenario when testing on the test scenario, new auxiliary tasks are introduced at the batch and sample level, and a new loss function is used to make the model better adapt to target data, so as to improve the accuracy of quality evaluation.

While existing deep learning-based NR-IQA methods have achieved promising performance, most of them are trained with single-quality labels. Hence, it would be challenging for these models to find an accurate mapping between features and

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.

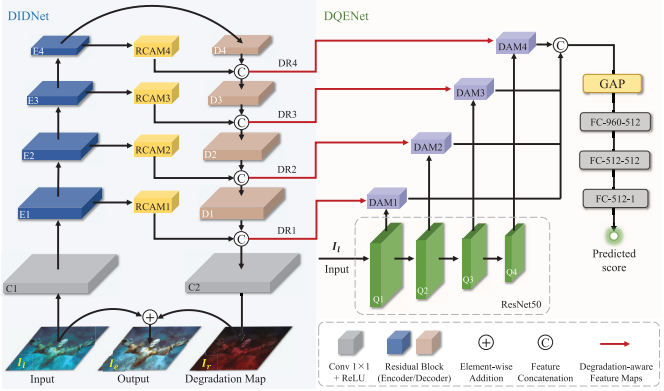


Fig. 3. **Overall architecture.** Our EDANet contains two parts including the DIDNet for exploiting degradation-aware information and the DQENet for quality prediction. These two parts have connections where the intermediate feature representations from the decoder side of the DIDNet are embedded into the DQENet to enhance the feature characterization capability. Our EDANet is the first of its kind to output both a degradation map and a single quality score, providing quantifiable means to better solve the challenging UIQA problem.

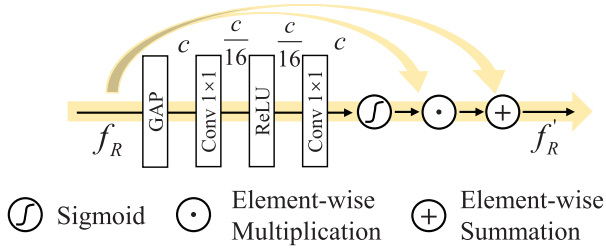


Fig. 4. **Architecture of RCAM**. Our RCAM serves as a skip connection between the encoder and decoder. It takes the encoded features as input and attentively highlights the most informative clues among channel dimensions, leading to better awareness of the degradation patterns. We provide the channel number changes in the figure to support better understanding of our operations.

quality labels, especially in those cases where a large variety of distortion types exist. By contrast, in this paper, we propose to leverage the detailed degradation-aware intermediate representations as supplements to guide the quality assessment process. By experiments, we show that, compared with the models which are trained with single-quality labels, our proposed EDANet with degradation-aware information embedding can achieve significantly better performance.

III. Methodology

A. Overview

The framework of our proposed EDANet is shown in Fig. 3. It contains two parts including the DIDNet for exploiting degradation-aware information and the DQENet for quality prediction. These two parts have connections where the intermediate feature representations from the decoder side of the DIDNet are embedded into the DQENet to enhance the feature characterization capability. The DIDNet and DQENet are separately trainable and we adopt a two-stage training strategy. Specifically, in the first stage, we train a DIDNet for residual learning on existing UIE dataset to fully exploit the degradation characteristics of underwater images. In the second stage, we freeze the model parameters of the DIDNet and leverage the multilevel degradation-aware feature representations from the decoder side of the DIDNet to assist in the training of the DQENet. Compared to one-step training, EDANet employs a two-stage training strategy which enables distinct specialization of DIDNet and DQENet, optimizing each network's performance in degradation analysis and quality prediction, respectively. Using intermediate degradation-aware feature representations as supplementary guidance, our proposed EDANet can more effectively evaluate the quality of underwater images. In the following, we will illustrate the details of DIDNet and DQENet, respectively. Finally, the loss functions are presented.

B. Architecture of DIDNet

The DIDNet takes low-quality underwater images as input and aims to enhance them through residual learning. In other words, the goal is to enhance the input underwater image by inferring a residual map between the input underwater image and its corresponding ground truth. The process is described as follows:

$$I_e = I_r + I_l = F_{DIDNet}(I_l; W_{DIDNet}) + I_l, \tag{1}$$

where I_e denotes the enhanced result of the input underwater image I_l, F_{DIDNet} denotes the DIDNet to generate the corresponding residual map I_r with W_{DIDNet} as the network parameters. Denote the ground truth high-quality version of I_l as I_p, the training target of F_{DIDNet} is to find the optimal parameter setting $\hat{W}_{DIDNet}$ by minimizing the loss between I_e and I_p over all training samples.

The DIDNet follows a classical U-Net-like encoder-decoder architecture. U-Net is famous for its ability to capture global and local features in an image. Meanwhile, it introduces the concept of skip connection, i.e., the feature maps of some layers in the encoder are directly connected to the corresponding layer of the decoder, so that the decoder can use feature information from different scales, which helps to improve the model's ability to capture details and local features. For the task of generating the residual degradation map, the network needs to accurately capture the degradation features of the image, including not only the global features, but also the detailed and local features. Therefore, the choice of U-Net-like model as the generation model matches well with the task of generating degraded residual maps from underwater images. However, instead of using cascaded convolution layers [56], [57], [58], we resort to the residual block [59] as the basic unit to build our backbone. Our motivation comes from the observation that the cascaded design may lead to the loss of content information during gradient propagation. Using the residual block with an additional skip connection will enable one to better transfer information through the deep network and better retrieve the content information from both low-level and high-level features to model the target degraded information. Besides, we propose an additional attention mechanism for the U-Net-like backbone. Specifically, we introduce a Residual Channel-wise Attention Module (RCAM) which takes the encoded features as input and attentively highlights the most informative clues among channel dimensions, leading to better awareness of the degradation patterns. Then, the attentive multilevel feature representations (processed by the RCAM

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.

TABLE I

FEATURE SIZE OF EACH MODULE IN THE DIDNET, WHERE $H \times W \times C$ REPRESENTS THE HEIGHT, WIDTH, AND CHANNEL NUMBER OF THE FEATURE MAP, RESPECTIVELY

Module	Layer	Input Size	Output Size
Encoder	C1	$H \times W \times 3$	$H/2 \times W/2 \times 32$
	E1	$H/2 \times W/2 \times 32$	$H/2 \times W/2 \times 32$
	E2	$H/2 \times W/2 \times 32$	$H/4 \times W/4 \times 64$
	E3	$H/4 \times W/4 \times 64$	$H/8 \times W/8 \times 128$
	E4	$H/8 \times W/8 \times 128$	$H/16 \times W/16 \times 256$
RCAM	RCAM1	$H/2 \times W/2 \times 32$	$H/2 \times W/2 \times 32$
	RCAM2	$H/4 \times W/4 \times 64$	$H/4 \times W/4 \times 64$
	RCAM3	$H/8 \times W/8 \times 128$	$H/8 \times W/8 \times 128$
	RCAM4	$H/16 \times W/16 \times 256$	$H/16 \times W/16 \times 256$
Decoder	D1	$H/16 \times W/16 \times 256$	$H/16 \times W/16 \times 256$
	D2	$H/16 \times W/16 \times 512$	$H/8 \times W/8 \times 128$
	D3	$H/8 \times W/8 \times 256$	$H/4 \times W/4 \times 64$
	D4	$H/4 \times W/4 \times 128$	$H/2 \times W/2 \times 32$
	C2	$H/2 \times W/2 \times 64$	$H \times W \times 3$

module) are concatenated with the corresponding feature representations in the decoder side. The details of RCAM are shown in Fig. 4. Specifically, let f_R be the input of our RCAM block. We first squeeze the spatial resolution by applying the global average pooling (GAP) on f_R. Then, we adopt two 1×1 convolutions for feature modeling, followed by ReLU and Sigmoid activation, respectively. This step aims to compute the overall semantic attention with a strong regularization along the channel direction. Then the obtained feature representation is multiplied by the input feature to calibrate the contribution of each channel. Note that our channel attention differs from that of the conventional one in that our computational cost is significantly lower due to the two cascaded 1×1 convolution layers (to reduce the number of channels). Finally, we adopt a residual addition to facilitate feature propagation. The whole RCAM can be formulated as:

$$\begin{aligned} f'_R &= RCAM(f_R) \\ &= f_R + f_R \cdot (Conv_{1\times1}(Conv_{1\times1}(GAP(f_R)))). \end{aligned} \tag{2}$$

The final architecture of our DIDNet is shown in the left side of Fig. 3 and the detailed configuration is listed in Table I. Technically, our DIDNet can be regarded as an improved version of U-Net connection which contains an additional attention module, i.e., RCAM. Moreover, our architecture differs from most existing works with a U-Net-like architecture [60], [61], [62] on the fusion design. Specifically, the common practice is to adopt an early fusion, i.e., the enhanced features are first merged with hierarchical features and then fed into the decoder. Adversely, we adopt the late fusion architecture where the hierarchical features are first decoded and then merged with the enhanced features. Our objective is to make the output of each layer richer by containing features from both the encoding and decoding stages. Therefore, they can be better used to guide our subsequent DQENet.

To better understand the effectiveness of DIDNet, we provide several qualitative comparisons in Fig. 5. For the

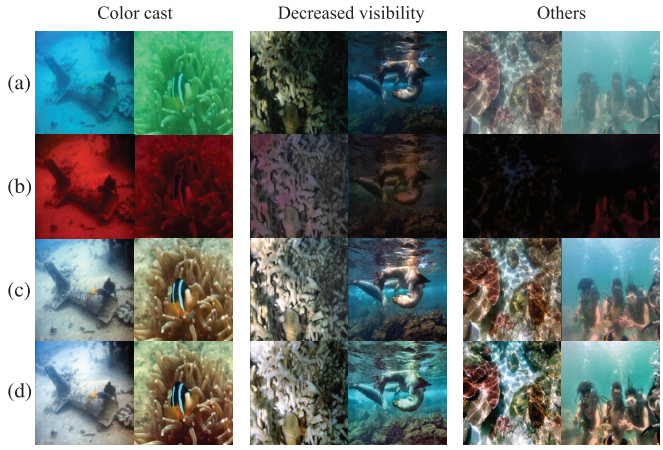


Fig. 5. **Visualization of the estimated residual maps and enhanced results by our DIDNet**. (a) Input low-quality images, (b) Residual maps, (c) Enhanced results, (d) Pseudo-reference images.

underwater scene, a number of images may suffer from a blue-green color bias during photography, as shown in the first two columns. In these cases, the residual map estimated by our DIDNet suggests that the image degradation mainly happens on the red channel, as shown in the 2nd row. This observation also correlates well with existing studies [63], [64]. By recovering from the degraded information, we can obtain an enhanced image (the 3rd row) with less color casts and higher visual quality, which is closer to the (pseudo) references shown in the 4th row. Another typical degradation type for the underwater scene is low visibility. As can be seen from the 3rd and 4th columns, the underwater images generally have low contrast and brightness due to light absorption. In such a case, our DIDNet can detect the degraded regions (as shown in the 2nd row) and successfully retrieve them, as can be observed by comparing the 3rd and 4th rows. More examples can be seen in the last two columns. Overall, the residual maps estimated by our DIDNet can well characterize the local degradation of underwater images and offer more intuitive image degradation information.

C. Architecture of DQENet

The architecture of our DQENet is shown on the right side of Fig. 3, and the detailed configuration is listed in Table II. Given an input underwater images, we first adopt the ResNet50 [59] as the backbone for feature extraction. Unlike most existing works that directly feed the extracted hierarchical features into a decoder for quality prediction, we explicitly embed the precomputed multilevel degradation-aware features in the decoder side of the DIDNet into our feature extraction module so as to best strengthen the degradation awareness of the extracted features. To achieve this, we propose a degradation-aware module (DAM) for feature embedding. The architecture of our DAM is shown in Fig. 6. Specifically, the DAM takes a pair of feature representations, including the quality feature representation f_Q and the degradation-aware feature representation f_D, as inputs. Then, a feature enhancement module (FEM) composed of a convolutional layer, an upsampling layer, and a convolution layer 1×1 is applied on

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.

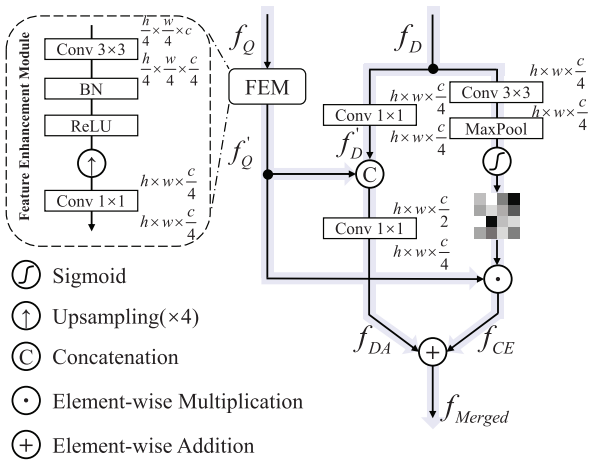


Fig. 6. **Architecture of DAM**. DAM takes the quality features f_Q and the degradation features f_D as input and fuses them to generate the degradation-aware feature f_{out}. Specifically, we first improve the feature modeling of f_Q through our FEM block by dealing with the channel and spatial dimension and form f'_Q. Meanwhile, the freezed-weight f_D is projected to a closer latent space to minimize the domain gap and form f'_D. Then, f'_Q and f'_D are fused together to generate the degradation-aware feature f_{DA}. Aside from this, we leverage the spatial clues from f_D to cross-enhance the f'_Q and form f_{CE}. The objective is to constrain the attention into locally-degraded regions. Lastly, f_{DA} and f_{CE} are merged together to generate the final output degradation-aware feature f_{Merged}. For ease of understanding, we provide the input and output feature sizes of each operation.

TABLE II

FEATURE SIZE OF EACH MODULE IN DQENET, WHERE $H \times W \times C$ REPRESENTS THE HEIGHT, WIDTH AND CHANNEL NUMBER OF THE FEATURE MAP, RESPECTIVELY

Layer	Input Size	Output Size
Q1	$H \times W \times 3$	$H/4 \times W/4 \times 256$
Q2	$H/4 \times W/4 \times 256$	$H/8 \times W/8 \times 512$
Q3	$H/8 \times W/8 \times 512$	$H/16 \times W/16 \times 1024$
Q4	$H/16 \times W/16 \times 1024$	$H/32 \times W/32 \times 2048$
DAM1	$f_Q : H/4 \times W/4 \times 256$ $f_D : H \times W \times 64$	$H \times W \times 64$
DAM2	$f_Q : H/8 \times W/8 \times 512$ $f_D : H/2 \times W/2 \times 128$	$H/2 \times W/2 \times 128$
DAM3	$f_Q : H/16 \times W/16 \times 1024$ $f_D : H/4 \times W/4 \times 256$	$H/4 \times W/4 \times 256$
DAM4	$f_Q : H/32 \times W/32 \times 2048$ $f_D : H/8 \times W/8 \times 512$	$H/8 \times W/8 \times 512$
FC1	960	512
FC2	512	512
FC3	512	1

the quality feature representation f_Q to adjust the resolution for matching that of the degradation feature representation f_D. Aside from this, the FEM module also contributes to mining the domain gap between the input features, i.e., the quality feature representation is projected in a latent space that is closer to that of the degradation feature representation. Mathematically, we can obtain the projected quality feature representation f'_Q by:

$$\begin{aligned} f'_Q &= FEM(f_Q) \\ &= Conv_{1\times1}(Upsample(Conv_{3\times3}(f_Q))). \end{aligned} \tag{3}$$

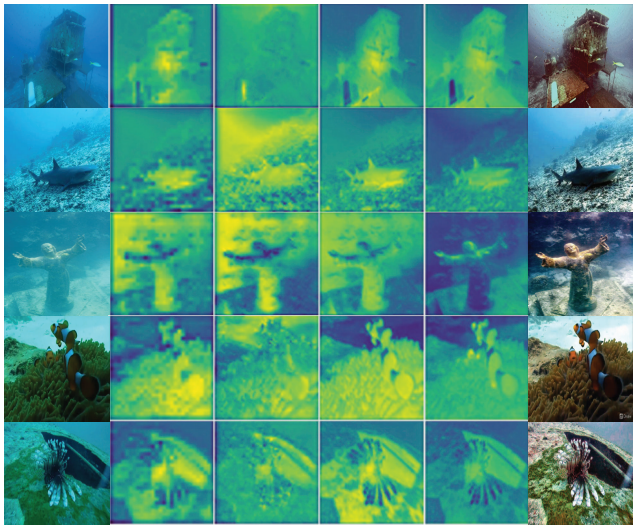


Fig. 7. **Degradation feature visualization of DIDNet**. (a) Input underwater images. (b)–(e) Multi-level degradation features. (f) Enhanced results.

Similarly, we also project the degradation feature representation f_D into the same latent space by applying a 1×1 convolution, forming the projected degradation feature representation f'_D. Then, f'_D is integrated with f'_Q by concatenation and convolution to derive the transited degradation-aware feature representation f_{DA}. Formally, we have:

$$f_{DA} = Conv_{1\times1}(Concat(Conv_{1\times1}(f_D), f'_Q)). \tag{4}$$

In addition, we also observed that the residual map estimated in the DIDNet can provide vital cues on local regions that are severely affected during underwater imaging. Intuitively, these regions are the key for the human visual system to perceive image quality. This observation motivates us to additionally use degradation-aware features as an extra clue to spatially guide the feature extraction process, which can help the network to pay more attention to those locally distorted regions. To achieve this goal, we extract the spatial attention map from f_D with the help of convolution, maxpooling, and sigmoid activation. Then, the spatial attention, referring to the degraded local regions, is applied to the projected quality feature f'_Q to form the cross-enhanced feature representation f_{CE}. Mathematically, we have:

$$f_{CE} = f'_Q \cdot (Conv_{3\times3}(f_D)). \tag{5}$$

Finally, the degradation-aware feature representation f_{DA} and the cross-enhanced feature representation f_{CE} are merged through addition to get the final output f_{Merged}:

$$f_{Merged} = f_{DA} + f_{CE}. \tag{6}$$

After applying the DAM at each level, we can obtain the merged feature representations at all levels, i.e., f^i_{Merged}, where i is the level index. The motivation of utilizing multi-level features comes from the observation that the degradation features at each level contain different information, as shown in Fig. 7(b)–(e). It can be seen that deep layer provides rough cues on the global degraded regions, while shallow layer contains more fine-grained details. Therefore, by leveraging

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.

features from all levels, our quality feature extraction process can benefit from both coarse and fine degradation-aware information to improve feature discriminability. Finally, we concatenate all the merged feature representations for quality prediction, i.e., feeding into the decoder composed of a GAP layer and three cascaded FC layers.

D. Network Training

We adopt a two-stage training strategy, i.e., the network parameters of the DIDNet and DQENet are optimized respectively. During the first stage, we only consider the DIDNet and train the network with full supervision. During the second stage, we freeze the pre-trained DIDNet and infer the degradation-aware feature representations in an off-the-shelf manner. Then, these prior knowledge on the degradation are adopted as the additional input to guide the training of the DQENet. Each subnetwork can be trained end-to-end.

1) DIDNet Training: Our DIDNet is trained on an existing UIE dataset, i.e., UIEB [65], which contains 890 raw underwater images along with their corresponding pseudo-reference images. We adopt two different kinds of losses during training: mean square error (MSE) loss $\mathcal{L}_{MSE}$ and perceptual loss $\mathcal{L}_{VGG}$. The $\mathcal{L}_{MSE}$ is a commonly used pixel-level loss function. Let I_e be the enhanced result and I_p be the pseudo reference. Formally, the MSE loss $\mathcal{L}_{MSE}$ is defined as follows:

$$\mathcal{L}_{MSE} = \|I_p - I_e\|_2^2, \tag{7}$$

where $\|\cdot\|_2$ represents the calculation of L_2 norm.

In addition to the pixel-wise constraint, we further leverage the perceptual loss to ensure consistency at the feature level. Specifically, we first apply a pre-trained VGG to extract features at different levels for both the enhanced result and the pseudo reference. Then, we compute their similarity values at these latent spaces. Formally, the perceptual loss $\mathcal{L}_{VGG}$ is defined as follows:

$$\mathcal{L}_{VGG} = \sum_{j=1}^{16} \frac{1}{H_j \times W_j \times C_j} \|\varphi_j(I_p) - \varphi_j(I_e)\|_2^2, \tag{8}$$

where $\varphi(\cdot)$ represents the pre-trained VGG-16; φ_j denotes the j_{th} layer of the VGG-16; H_j, W_j, and C_j are height, width, and channel of the output of the j-th layer of the VGG-16. Finally, the training of the DIDNet is supervised by weighting these two losses:

$$\mathcal{L}_D = (1-\alpha)\mathcal{L}_{MSE} + \alpha\mathcal{L}_{VGG}, \tag{9}$$

where α is the weighting hyperparameter. In our case, we set $\alpha = 0.1$ to emphasize much more importance on $\mathcal{L}_{MSE}$.

2) DQENet Training: Our DQENet is trained with paired underwater images along with their corresponding subjective quality scores. However, the form of quality score varies in different datasets. For example, for the conventional UWIQA dataset [47], the subjective quality score is provided in the form of an absolute quality score. Therefore, in such a case, we leverage the conventional mean absolute error (MAE) loss ($\mathcal{L}_{MAE}$) to measure the difference between the prediction quality score and the subjective quality score:

$$\mathcal{L}_{MAE}(y_i, \hat{y}_i) = \mid y_i - \hat{y}_i \mid, \tag{10}$$

where y_i represents the subjective quality score and $\hat{y}_i$ is the predicted quality score.

In the literature, a number of research [66] has revealed the benefits of pairwise comparison for subjective quality assessment. Indeed, the recently constructed SAUD dataset [48] is built with this idea and provides relative ranking scores instead of the absolute quality scores. Note that ranking scores are only meaningful for comparing the images within the same group. Therefore, while training on the SAUD dataset, we choose the binary preference label associated with a pair of homologous underwater images as supervision. Specifically, the pairwise ranking loss is adopted to ensure the consistency between the predicted ranking scores $\hat{Q}_i, \hat{Q}_j$ from two homologous images and those of the ground truth. The ranking loss is defined as follows:

$$\mathcal{L}_{Rank} = \max(0, -R \times (\hat{Q}_i - \hat{Q}_j) + \beta), \tag{11}$$

where $\hat{Q}_i$ and $\hat{Q}_j$ are the predicted ranking scores of the paired inputs; $R \in \{1, -1\}$ is the subjective binary preference. When $R = 1$, it means that the first image is subjectively preferred compared to the other. Otherwise, $R = -1$. β stands for the preference bias which is set to 0.5. When the difference between the predicted scores is greater than 0.5, the prediction is validated, otherwise the prediction is rejected.

IV. Experiments

A. Experimental Settings

1) UIQA Databases: We evaluate the performance of our approach on two public UIQA databases, i.e., SAUD [48] and UWIQA [47]. SAUD [48] contains 100 raw underwater images and 1,000 enhanced underwater images (enhanced by 10 UIE algorithms). A double stimuli-based pairwise comparison methodology is adopted for obtaining subjective quality labels. When conducting experiments on this database, 800 enhanced images derived from 80 raw images are selected for model training and the remaining 200 enhanced images from 20 raw images are used for testing. UWIQA [47] contains 890 images, each of which is labeled by 21 observers. The quality scores are ranged from 0.1 to 1. We also use 80% images (i.e., 712 images) for model training, and then the rest 20% images (i.e., 178 images) for testing.

2) Performance Criteria: We use the widely-acknowledged metrics, including Pearson linear correlation coefficient (PLCC), Spearman rank order correlation coefficient (SRCC), Kendall rank order correlation coefficient (KRCC), and root mean square error (RMSE), as the performance criteria. For each dataset, we run 10 times of random train-test dataset splitting and the median PLCC, SRCC, KRCC and PLCC values are reported.

3) Implementation Details: As for the pre-training of the DIDNet, we resort to the UIEB dataset [65] which contains 890 raw underwater images along with their corresponding pseudo-reference images. The UIEB dataset is constructed to be comprehensive, covering a wide range of underwater conditions, including various types of degradation such as color casts, reduced visibility, and blurring. This diversity ensures that DIDNet learns to recognize and process a broad spectrum

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE SAUD AND UWIQA DATABASES

Methods		SAUD [48]				UWIQA [47]				Weighted Average				Time(sec.)↓
		PLCC↑	SRCC↑	KRCC↑	RMSE↓	PLCC↑	SRCC↑	KRCC↑	RMSE↓	PLCC↑	SRCC↑	KRCC↑	RMSE↓	
Traditional NR-IQA	DIIVINE [16]	0.7598	0.6119	0.4818	12.8349	0.1206	0.4137	0.3486	0.1512	0.4588	0.5186	0.4191	6.8622	24.2272
	BRISQUE [17]	0.8150	0.7167	0.5884	11.0029	0.5595	0.5584	0.4304	0.1253	0.6947	0.6422	0.5140	5.8806	0.1419
	GM-LOG [18]	0.7929	0.6518	0.5128	11.8713	0.6314	0.6352	0.4918	0.1181	0.7168	0.6440	0.5029	6.3367	3.7614
	NFERM [19]	0.8020	0.6414	0.5143	11.6847	0.5278	0.5178	0.3962	0.1287	0.6733	0.5832	0.4587	6.2430	72.4397
	NRSL [20]	0.6661	0.6306	0.5034	12.8274	0.0719	0.0634	0.0459	0.1514	0.3863	0.3635	0.2880	6.8583	4.6096
	BIQME [21]	0.8383	0.7271	0.5924	10.7982	0.2874	0.4356	0.3200	0.1471	0.5789	0.5898	0.4641	5.7826	0.6790
Deep NR-IQA	CNN [25]	0.5260	0.5487	0.4222	13.7110	0.7211	0.7550	0.6026	0.1184	0.6179	0.6458	0.5072	7.3103	0.3344
	WaDIQaM-NR [26]	0.5405	0.6040	0.4738	13.1960	0.7635	0.7694	0.6202	0.0956	0.6455	0.6819	0.5427	7.0270	**0.0486**
	CaHDC [27]	0.3560	0.3790	0.2920	16.1846	0.3340	0.3310	0.2440	0.1425	0.3456	0.3564	0.2694	8.6304	0.3262
	DB-CNN [28]	0.8228	0.8347	0.6832	9.7498	0.7681	0.7858	0.6426	0.0968	0.7970	0.8117	0.6641	5.2042	0.0591
	VCRNet [29]	0.8175	0.8141	0.6632	9.9560	0.7865	0.7933	0.6469	0.0981	0.8029	0.8043	0.6555	5.3139	2.2001
	TReS [30]	0.8297	0.8002	0.6816	9.6377	0.6542	0.6508	0.5011	0.1495	0.7471	0.7298	0.5966	5.1697	2.5345
	TTDA [31]	0.7935	0.7675	0.5897	10.9222	0.7172	0.7323	0.5667	0.1915	0.7576	0.7509	0.5789	5.8691	2.1780
UIQA	UCIQE [44]	0.7219	0.5184	0.3863	14.8405	0.5749	0.5844	0.4446	0.1246	0.6527	0.5495	0.4138	7.9108	0.2207
	UIQM [45]	0.6957	0.4085	0.3098	14.1798	0.2074	0.2964	0.2158	0.1490	0.4658	0.3557	0.2655	7.5727	2.1592
	CCF [46]	0.0157	0.0443	0.0341	15.5915	0.4471	0.4426	0.3309	0.1363	0.2188	0.2319	0.1739	8.3137	1.2336
	FDUM [47]	0.2095	0.1622	0.1047	16.3623	0.6406	0.6761	0.5256	0.1170	0.4125	0.4042	0.3029	8.7124	2.8991
	NUIQ [48]	0.8679	0.7900	0.6555	9.7666	0.7214	0.6948	0.5497	0.1047	0.7989	0.7452	0.6057	5.2168	1.5663
	EDANet	**0.8792**	**0.8517**	**0.7459**	**9.3911**	**0.8156**	**0.8104**	**0.6661**	**0.0860**	**0.8493**	**0.8323**	**0.7083**	**5.0093**	0.0768

of degradation characteristics inherent to underwater imagery. The size of raw underwater images and their corresponding pseudo reference images are resized to 224×224. The learning rate is set to $1e{-}4$, and the Adam optimizer is used to minimize the loss function for 200 epochs. The batch size is set to 10. For each epoch, we trained with the training portion of the entire dataset. Then, the parameters of the DIDNet are frozen during the training of DQENet.

As for the training of DQENet, the size of images is also resized to 224×224. We also use the Adam optimizer for minimizing the loss function for 50 epochs, and the learning rate is set to $1e-4$. The batch size is set to 10. The above networks are both implemented by Pytorch and trained with an NVIDIA 2080Ti GPU.

B. Experimental Results

Since there is always no available pristine reference for UIQA, we compare our methods against 18 existing NR-IQA methods, including six hand-craft feature-based NR-IQA methods (i.e., DIIVINE [16], BRISQUE [17], GM-LOG [18], NFERM [19], NRSL [20], and BIQME [21]), seven deep NR-IQA methods (i.e., CNN [25], WaDIQaM-NR [26], CaHDC [27], DB-CNN [28], VCRNet [29], TReS [30], and TTDA [31]), and five UIQA methods (i.e., UCIQE [44], UIQM [45], CCF [46], FDUM [47], and NUIQ [48]). The hand-craft feature-based NR-IQA methods and UIQA method include two types: training-based and training-free. Among them, CCF and FDUM are training-free (i.e., no retraining is required) while the rests are all retrained on the SAUD database and the UWIQA database, respectively. For the retraining, the regression tool is implemented by SVMrank on SAUD and SVR on UWIQA, respectively. By contrast, the learning-based methods directly optimize an end-to-end function mapping from the input image to its quality score while without requiring any effort on manual feature engineering.

1) Performance on individual database: To evaluate the effectiveness of the proposed EDANet, the quantitative comparison results of different NR-IQA methods on the SAUD and UWIQA databases are shown in Table III. It should be noted that we are more concerned with the weighted average performance improvement of different methods on the SAUD datasets and UWIQA datasets. Therefore, we additionally provide the weighted average performance results of different methods on the two datasets. The weights are set according to the number of image samples contained in each dataset. As can be seen from the table, we can conclude that the superiority of our method is four-fold.

First, our method outperforms all the competing methods in both databases. The results also clearly show that our method has a significant performance improvement in terms of weighted average performance on the two datasets. Existing UIQA methods like FDUM, CCF, and UIQM struggle to accurately predict the quality of underwater images in challenging scenarios, such as severe color shifts or significantly reduced visibility. These UIQA methods evaluate the quality of underwater images from specific aspects, such as color, contrast, and clarity. In contrast, instead of measuring specific distortion levels in an explicit manner, our proposed EDANet exploits general degradation-aware information, which encompasses a more comprehensive degradation characteristics to facilitate quality prediction. To the best of our knowledge, this is the first time that extra explicit degradation-sensitive information is used to guide quality-sensitive feature learning in the task of UIQA.

Second, most training-based methods perform better than training-free ones (i.e., CCF and FDUM) while the deep learning-based methods are superior to most hand-craft

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE TWO SUBSETS OF SAUD DATABASE

Methods		Subset-I (Biased Color Subset)				Subset-II (Reduced Visibility Subset)			
		PLCC↑	SRCC↑	KRCC↑	RMSE↓	PLCC↑	SRCC↑	KRCC↑	RMSE↓
Traditional NR-IQA	DIIVINE [16]	0.7224	0.5538	0.4321	13.6478	0.8162	0.7038	0.5698	11.9118
	BRISQUE [17]	0.7820	0.6716	0.5398	11.8452	0.8798	0.7902	0.6722	9.7639
	GM-LOG [18]	0.7725	0.5998	0.4628	12.2204	0.8407	0.7320	0.5999	11.0313
	NFERM [19]	0.7611	0.5864	0.4638	11.0139	0.8444	0.7388	0.6045	12.5561
	NRSL [20]	0.7527	0.6150	0.4856	12.9600	0.8087	0.6661	0.5358	12.3593
	BIQME [21]	0.8235	0.6993	0.5629	10.9177	0.8613	0.7655	0.6310	10.3716
Deep NR-IQA	CNN [25]	0.4931	0.4795	0.3834	14.1490	0.6196	0.6159	0.5048	12.8453
	WaDIQaM-NR [26]	0.4767	0.5665	0.4357	12.4201	0.6770	0.6931	0.5382	11.5964
	CaHDC [27]	0.3550	0.3480	0.2650	15.4280	0.3580	0.4260	0.3330	14.6890
	DB-CNN [28]	0.7656	0.7761	0.6581	10.3796	0.8354	0.8597	0.7465	9.6253
	VCRNet [29]	0.8286	0.7787	0.6549	10.1283	0.8485	0.8349	0.7018	10.0252
	TReS [30]	0.8145	0.7938	0.6805	10.0568	0.8315	0.8078	0.6837	9.5915
	TTDA [31]	0.7762	0.7537	0.5762	11.2518	0.8105	0.7842	0.6016	10.1135
UIQA	UCIQE [44]	0.6985	0.4487	0.3317	14.5375	0.7598	0.6162	0.4684	13.7276
	UIQM [45]	0.6748	0.4004	0.2938	13.8795	0.7142	0.4711	0.3619	14.2930
	CCF [46]	0.0909	0.0937	0.0629	15.5222	0.0972	0.0298	0.0090	15.1955
	FDUM [47]	0.1510	0.1392	0.0940	15.6259	0.2973	0.1965	0.1206	15.3047
	NUIQ [48]	0.8610	0.7808	0.6426	9.7099	0.8760	0.7973	0.6759	9.5994
	EDANet	**0.8755**	**0.8450**	**0.7373**	**9.3489**	**0.8818**	**0.8630**	**0.7601**	**9.4502**

feature-based methods. This phenomenon is reasonable because UIQA is a challenging task in which training is particularly useful to model the complex non-linear relationship between extracted features and perceived quality score. The end-to-end deep learning technique further provides an effective solution to directly establish explicit image-to-quality mapping. The superiority of performance is also one of the important reasons that deep learning is widely used in many IQA tasks. As part of the deep learning-based method, our network sets new records on both datasets in terms of all metrics.

Third, it can be seen that several IQA methods like DIIVINE, BRISQUE, and NRSL perform well on one dataset while they do not work well on another. This may be due to the fact that the content and degradation types of underwater images contained in the two datasets are not exactly the same. Therefore, these methods cannot generalize well across datasets. Meanwhile, it can be seen that our method can consistently outperform all the counterparts on both datasets, validating the effectiveness and robustness of our network. Several recent learning-based methods, especially the recent NUIQ, also perform well on both datasets. However, it can be seen that, in terms of model complexity, our EDANet requires significantly less time for inference, i.e., nearly 20 times faster than NUIQ.

Fourth, UIQA methods such as UCIQE and UIQM, which are specifically developed for underwater images, often find themselves inferior to deep-learning techniques like DB-CNN and VCRNet, which are initially designed for natural imagery. This discrepancy becomes evident especially after retraining these deep-learning models on specialized UIQA databases like SAUD and UWIQA. These models, through extensive pre-training, establish a solid foundation that, after undergoing several rounds of fine-tuning for underwater-specific tasks, allows them to adapt and perform effectively in the UIQA task.

2) Performance on individual distortion types: To verify the performance of the proposed EDANet on different distortion types, we provide detailed performance results of dealing with specific distortions. The quantitative results are shown in Table IV. It is observed that nearly all methods achieve better performance on the reduced visibility subset compared to the biased color subset. This trend can mainly be ascribed to the ability of most IQA methods to address structural features rather than color biases. This pattern underscores the current IQA approaches' emphasis on maintaining structural integrity at the expense of color accuracy, pointing to a significant area for development in techniques sensitive to color assessment. The observed phenomenon also indicates that color cast is a primary bottleneck for UIQA methods. Given the underwater environment's impact, most underwater images exhibit color casts, resulting in a biased color appearance and posing a significant challenge for UIQA. However, our EDANet delivers consistent superior performance on both subsets. Compared to reduced visibility, the biased color is recognized as one of the foremost challenges in the context of UIQA. Our proposed method stands out for its consistent and superior performance, particularly in handling images within the biased color subset. This is largely because our proposed EDANet incorporates more refined degradation-aware information, especially tailored for identifying and addressing color casts. The effectiveness of our method is further illustrated through the visualization of residual maps between the input raw underwater images and the enhanced results by our DIDNet. These residual maps serve as a testament of DIDNet's capability to accurately discern the extent of color cast degradation, as shown in the first two columns of Fig. 5. These features are subsequently integrated into our DQENet, enhancing its ability to predict image quality with a focus on color degradation. Essentially, by embedding

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.

TABLE V
PREDICTION ACCURACY OF THE NETWORK WITH AND WITHOUT THE PROPOSED DIDNET

Ablation Models	SAUD [48]				UWIQA [47]			
	PLCC↑	SRCC↑	KRCC↑	RMSE↓	PLCC↑	SRCC↑	KRCC↑	RMSE↓
w/o DIDNet	0.8134	0.8055	0.6931	10.3146	0.7436	0.7667	0.6194	0.1124
w DIDNet	**0.8792**	**0.8517**	**0.7459**	**9.3911**	**0.8156**	**0.8104**	**0.6661**	**0.0860**

TABLE VI
ABLATION STUDIES ON THE PROPOSED EDANET

DIDNet	Skip Module	Conventional [67]	RCAM	RCAM	RCAM	RCAM	RCAM
	Hyperparameter α	0.1	1	0.05	0.1	0.1	0.1
	Early/Late Fusion	Late	Late	Late	Early [60]	Late	Late
DQENet	Loss Type	$\mathcal{L}_{Rank}$	$\mathcal{L}_{Rank}$	$\mathcal{L}_{Rank}$	$\mathcal{L}_{Rank}$	$\mathcal{L}_{MAE}$	$\mathcal{L}_{Rank}$
Criteria	PSNR (dB)↑	21.8353	19.6773	21.5210	21.9424	/	**22.3528**
	SSIM↑	0.9129	0.8703	0.9058	0.9214	/	**0.9306**
	MSE ($\times 10^3$)↓	0.6193	1.1399	0.6204	0.5669	/	**0.5484**
	PLCC↑	/	/	/	/	0.8416	**0.8792**
	SRCC↑	/	/	/	/	0.8446	**0.8517**
	KRCC↑	/	/	/	/	0.7044	**0.7459**
	RMSE↓	/	/	/	/	9.6274	**9.3911**

detailed degradation-aware information—particularly regarding color casts—into our quality evaluation network, a robust performance across both subsets can be achieved.

V. ABLATION STUDIES

A. Analysis on the DIDNet

Qualitatively, to validate the effectiveness of DIDNet, we have provided the estimated residual maps as well as the generated feature maps in Fig. 5 and Fig. 7, respectively. It can be observed that the estimated residual maps by our DIDNet can effectively detect the of underwater image degradation regions as well as the degradation degrees. In this subsection, we conduct a quantitative ablation study on the DIDNet and analyze the influence on the overall performance of quality prediction. The experimental results are listed in Table V where "w/o DIDNet" denotes the performance without using the degradation-aware information from the DIDNet, while "w DIDNet" represents the DIDNet is retained, i.e., the complete EDANet. It can be seen that, with the help of DIDNet, the accuracy of quality prediction can be significantly improved.

Besides, our DIDNet borrows the architecture of the U-Net but is different from U-Net which has been mentioned in Section III-B. Specifically, our DIDNet fuses the feature maps of encoder and decoder in a late fusion manner, which is different from the fusion scheme of early fusion in traditional U-Net architecture. Therefore, ablation studies are performed to demonstrate the effectiveness of our proposed DIDNet and the results are shown in the 6th and the 8th column of Table VI. As we can observe, compared to the conventional U-Net-like architecture, our proposed DIDNet (with the late fusion manner) has delivered better performance quantitatively.

We also provide detailed ablation studies on the used hyperparameter (Eq. 9) which weighs the relative contributions of $\mathcal{L}_{MSE}$ and $\mathcal{L}_{VGG}$. Table VI shows the PSNR, SSIM, and MSE results achieved by setting the hyperparameter α to 1, 0.1, and 0.05, respectively. It can be observed that with the hyperparameter of $\alpha = 0.1$, the best performance can be achieved. It is also worth noting that our DIDNet performs generally well and is not highly sensitive to hyperparameter.



Fig. 8. **Qualitative comparison on the RCAM block.** With the help of the RCAM, our network can output recovered images closer to the pseudo-reference image. (a) Input image, (b) w/o RCAM, (c) w RCAM, (d) Pseudo-reference image.

B. Analysis on RCAM

The RCAM is adopted in the DIDNet as a kind of channel attention to filter the encoded features. To prove the efficiency of the proposed RCAM, we first qualitatively compare the visual restoration performance of the DIDNet with and without the RCAM. The results are shown in Fig. 8, in which the "w/o RCAM" indicates that the degradation maps and the enhanced results are generated without the RCAM, and the "w RCAM" donates that the degradation maps and the enhanced results are generated with the RCAM. It can be observed that without the RCAM, the network can enhance the input image, but the estimated degradation map is more blurry, i.e., the contour of degraded regions/objects is less clear. In other words, the capabilities of color correction and detail recovery are decreased. On the contrary, with the help of the RCAM, the

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.

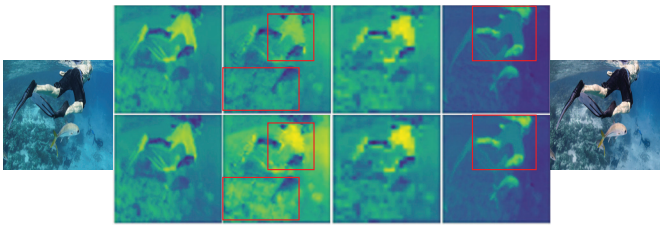


Fig. 9. **Feature visualization before and after the RCAM block**. With the help of the RCAM, we enable feature modeling with more activation on the regions of interest, highlighted by the **red** box. (a) Input image, (b-e) Multi-level features in the encoder, (f) Pseudo-reference image.

DIDNet can obtain much better enhanced results, i.e., closer to the pseudo reference images.

To better understand the utility of our RCAM, we provide the feature visualization before and after the RCAM block in Fig. 9. Fig. 9 (b)-(e) stand for the multi-scale encoded features. The first row visualizes the features before RCAM, while the second row visualizes the enhanced features after RCAM. Through comparison, it can be found that the RCAM can improve the feature modeling with more activation on the regions of interest, highlighted by the red box. We can also observe that the enhanced features can provide clearer edges with less noise, which shows that our RCAM can efficiently enhance the multi-level information from the encoder and propagate it to the decoder. Thus, the network can better characterize both global and local degradations.

In addition, our channel attention differs from the conventional one. Therefore, a comparison of the conventional channel attention and the proposed RCAM is also conducted. The results are shown in the 3rd and the 8th column of Table VI. It can be seen that our proposed RCAM has improved the performance of image enhancement, with higher PSNR and SSIM values while lower RMSE values.

C. Analysis on Quality Loss

As illustrated in Eq. 11, we choose the $\mathcal{L}_{Rank}$ to supervise the training of our DQENet. To validate the effectiveness of this choice, we conduct ablation experiments by replacing the $\mathcal{L}_{Rank}$ with the conventional $\mathcal{L}_{MAE}$ [68], while maintaining all the other training settings. The quantitative comparison can be found in the 7th and the 8th column of Table VI. We can observe that the adopted $\mathcal{L}_{Rank}$ can significantly improve the model accuracy in terms of all performance metrics.

D. Model Complexity on DIDNet

Our EDANet contains an additional DIDNet which aims to exploit degradation-aware information by pre-training a DIDNet for residual learning on the existing UIE datasets. To make the whole inference stage lightweight, we particularly and carefully design a tailored network, i.e., DIDNet. The comparison against other deep UIE methods can be found in Table VII. We select DenseGAN [69], Water-Net [65], and Ucolor [39] for comparison because our paper focuses on UIQA instead of UIE itself, and the design of the UIE module is intended to extract degradation-aware features to better guide quality assessment. However, the additional UIE module will inevitably increase computational costs, and our goal is to design a UIE module with low computational overhead. These compared methods, i.e., Dense-GAN, WaterNet, and Ucolor have shown outstanding performance in the UIE task and are widely used. Among them, WaterNet is well recognized (has very high citation number) and has relatively low computational complexity. Our experiment results also demonstrate that our DIDNet requires the least FLOPs, proving the effectiveness of our proposed DIDNet in guiding UIQA with relatively low complexity. With the help of the degradation features computed from such a lightweight network, our whole EDANet can achieve the inference speed with 0.076 second per image as shown in Table III. With the high inference speed, our EDANet has a great potential for real-time systems.

TABLE VII
QUANTITATIVE COMPARISON RESULTS OF DIDNET AND OTHER REPRESENTATIVE UIE METHODS ON THE UIEB DATASET

Method	Dense GAN	Water-Net	Ucolor	DIDNet
PSNR (dB)↑	17.2843	19.1130	20.6300	**22.3528**
SSIM↑	0.4426	0.7971	0.8239	**0.9306**
MSE ($\times 10^3$)↓	1.2152	0.7976	0.7700	**0.5484**
Params (MB)↓	/	**1.09**	148.77	70.653
FLOPS (G)↓	/	571.61	11219.78	**14.4017**

E. Analysis on the DAM

The DAM is adopted to fuse the multi-level degradation-aware feature maps from the DIDNet and the multi-level feature maps from the feature extraction backbone of DQENet. Specifically, it implements a weighted fusion on the quality feature maps with the corresponding degradation-aware feature maps. In this way, the training of the DQENet is guided by the pre-computed degradation-aware knowledge. To verify the efficiency of the proposed DAM, we simply replace the DAM with the simple concatenation operation. The experimental results are listed in Table VIII, in which the network without using the DAM is donated as "w/o DAM," and "w DAM" represents the DAM retained. It can be seen that the DAM can effectively fuse the quality features with the intermediate degradation features to achieve a more accurate prediction.

F. Coupling Degradation Maps With Quality Scores

One major advantage of our EDANet is the capability to output a detailed degradation map identifying the local distorted regions as well as a single quality score indicating the overall distortion level by the two sub-networks, i.e., DIDNet and DQENet, respectively. We argue that there exists an inner correlation between them which has been validated through previous and exhaustive experiments. To further validate our assumption and better support the design of our network, we present a kind of visualizations in Fig. 10. From the input images with the associated MOS values, we can observe that images with severe color casts tend to have lower quality scores. This correlation is successfully learned by our EDANet since it can accurately predict quality scores and the generated degradation maps can well reflect the degradation degrees of

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
QUALITY PREDICTION PERFORMANCE OF THE NETWORK WITH AND WITHOUT THE PROPOSED DAM

Ablation Models	SAUD [48]				UWIQA [47]			
	PLCC↑	SRCC↑	KRCC↑	RMSE↓	PLCC↑	SRCC↑	KRCC↑	RMSE↓
w/o DAM	0.8251	0.8157	0.6966	9.9555	0.7964	0.7964	0.6431	0.0912
w DAM	**0.8792**	**0.8517**	**0.7459**	**9.3911**	**0.8156**	**0.8104**	**0.6661**	**0.0860**

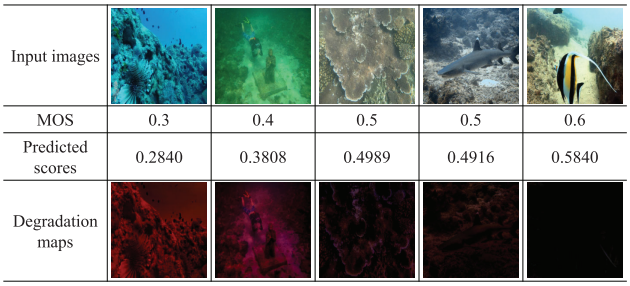


Fig. 10. **Relationship between degradation map and quality score.** From left to right, the quality of the input underwater image becomes better, i.e., with less color casts and with higher MOSs. The degradation maps estimated by our DIDNet become less visually noticeable. This phenomena indicates that our EDANet can effectively benefit from the degradation-aware representations to facilitate image quality prediction.

red channels. When the color cast is not obvious, our generated degradation maps also become less noticeable. These results show that our DIDNet can effectively model the inner relationship between degradation information and image quality.

In addition, we also simulate two common distortions of underwater images, i.e., color casts and visibility reduction, on a high-quality underwater image. To simulate color casts, we reduce the image pixel intensity values in the red channel by 40, 80, 120, and 160, respectively. To simulate visibility reduction, we reduce the pixel intensity value in all image channels by 20%, 40%, 60%, and 80%, respectively. These simulated lower-quality images can be found in Fig. 11. Then, we feed these simulated images into our EDANet. It can be seen that, when we increase the distortion degree, our EDANet can not only effectively predict lower scores as expected, but also generate more visible degradation maps with severer visual distortions. In Fig. 11, we also visualize the prediction error maps between the generated residual maps and the ground truth residual maps to demonstrate the effectiveness of our generated residual maps by DIDNet. It can be observed that these error maps are biased toward the dark (with small error), which strongly indicates the effectiveness and accuracy of the residual maps generated by our DIDNet. Through this experiment, we further validate that our EDANet has successfully learned the implicit link between the degradation information and the quality score. Thus, the rationale for exploiting the degradation-aware information first and then embedding such information to promote quality prediction is well demonstrated.

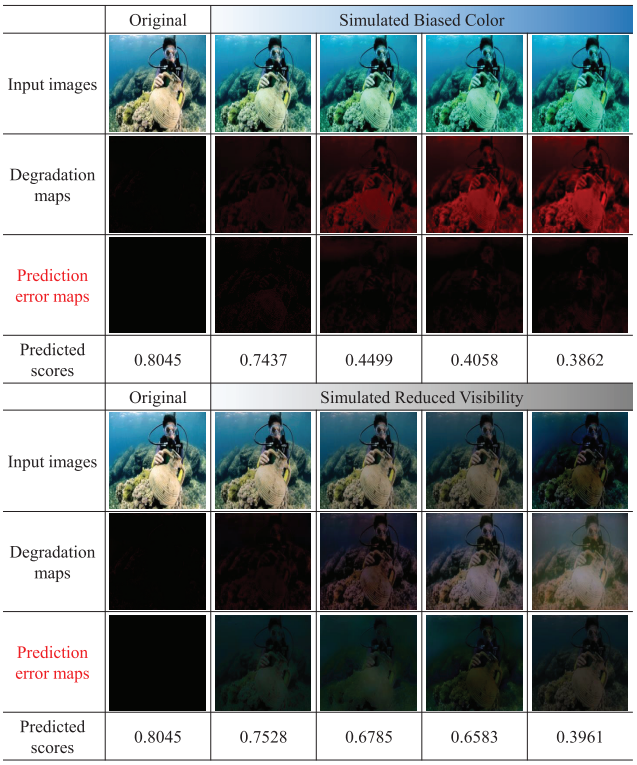


Fig. 11. **The predicted quality scores and generated degradation maps by our EDANet of the underwater images with simulated distortions**. By simulating different levels of degradations, our network can always consistently generate the degradation maps and output reasonable quality scores.

G. Discussions

Although the effectiveness of our method is demonstrated primarily within the context of UIQA, it addresses specific challenges such as color casts and reduced visibility. However, the underlying principles of embedding degradation-aware information into the quality evaluation process hold potential applicability beyond just underwater imagery. In fact, it is plausible that similar approaches could be developed to evaluate the quality of other types of images where color distortion, visibility reduction, and other forms of degradation are prevalent. For example, images captured in foggy or smoky conditions, or those affected by lighting variations, might benefit from the methodology applied in our study. Future investigations could extend the application of this method to these relevant scenarios, adjusting and optimizing the degradation-aware feature extraction and quality prediction components to suit different types of image degradation. Such explorations would not only validate the versatility of the proposed approach, but also potentially broaden its utility across a wider range of image processing and quality assessment tasks.

VI. CONCLUSION

In this paper, we have presented a novel UIQA model based on explicit degradation awareness embedding called EDANet. The core idea is to exploit informative degradation-aware features to guide the quality prediction process. The

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.

EDANet consists of two subnetworks including a DIDNet for degradation-aware feature generation and a DQENet for quality prediction. These two subnetworks are separately end-to-end trainable, and we adopt a two-stage training strategy. In the first stage, a DIDNet is pre-trained to estimate the residual map between the input degraded underwater image and its corresponding pseudoreference counterpart. In the second stage, we integrate the degradation-aware features acquired by our DIDNet into our DQENet to guide the end-to-end learning process. Unlike previous works, our EDANet is the first of its kind to output a local degradation map as well as a single quality score. In such a manner, we can provide quantifiable means to better understand visual quality, which is the first time achieved in UIQA. The effectiveness of our proposed EDANet has been validated through extensive experiments on two publicly available UIQA databases. The underlying principles of embedding degradation-aware information into the quality evaluation process hold potential applicability beyond just underwater imagery. In future work, one can extend the application of our method to other relevant scenarios.

REFERENCES

[1] M. Ludvigsen, B. Sortland, G. Johnsen, and H. Singh, "Applications of geo-referenced underwater photo mosaics in marine biology and archaeology," *Oceanography*, vol. 20, no. 4, pp. 140–149, Dec. 2007.

[2] M. Yang, J. Hu, C. Li, G. Rohde, Y. Du, and K. Hu, "An in-depth survey of underwater image enhancement and restoration," *IEEE Access*, vol. 7, pp. 123638–123657, 2019.

[3] B. L. Mcglamery, "A computer model for underwater camera systems," in *Proc. Soc. Photo-Org. Instrum. Engineers (SPIE) Conf. Ser.*, vol. 208, 1980, pp. 221–231.

[4] J. S. Jaffe, "Computer modeling and the design of optimal underwater imaging systems," *IEEE J. Ocean. Eng.*, vol. 15, no. 2, pp. 101–111, Apr. 1990.

[5] W. Hou, S. Woods, E. Jarosz, W. Goode, and A. Weidemann, "Optical turbulence on underwater image degradation in natural environments," *Appl. Opt.*, vol. 51, no. 14, p. 2678, 2012.

[6] D. Akkaynak and T. Treibitz, "A revised underwater image formation model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6723–6732.

[7] D. Akkaynak, T. Treibitz, T. Shlesinger, Y. Loya, R. Tamir, and D. Iluz, "What is the space of attenuation coefficients in underwater computer vision?" in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 568–577.

[8] Y. Kang, Q. Jiang, C. Li, W. Ren, H. Liu, and P. Wang, "A perception-aware decomposition and fusion framework for underwater image enhancement," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 3, pp. 988–1002, Mar. 2023.

[9] Q. Jiang, Y. Mao, R. Cong, W. Ren, C. Huang, and F. Shao, "Unsupervised decomposition and correction network for low-light image enhancement," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 19440–19455, Oct. 2022.

[10] W. Zhou, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, pp. 600–612, 2004.

[11] H. R. Sheikh and A. C. Bovik, "Image information and visual quality," *IEEE Trans. Image Process.*, vol. 15, no. 2, pp. 430–444, Feb. 2006.

[12] L. Zhang, Y. Shen, and H. Li, "VSI: A visual saliency-induced index for perceptual image quality assessment," *IEEE Trans. Image Process.*, vol. 23, no. 10, pp. 4270–4281, Oct. 2014.

[13] W. Xue, L. Zhang, X. Mou, and A. C. Bovik, "Gradient magnitude similarity deviation: A highly efficient perceptual image quality index," *IEEE Trans. Image Process.*, vol. 23, no. 2, pp. 684–695, Feb. 2014.

[14] K. Ding, K. Ma, S. Wang, and E. P. Simoncelli, "Image quality assessment: Unifying structure and texture similarity," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2567–2581, May 2020.

[15] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 586–595.

[16] A. K. Moorthy and A. C. Bovik, "Blind image quality assessment: From natural scene statistics to perceptual quality," *IEEE Trans. Image Process.*, vol. 20, no. 12, pp. 3350–3364, Dec. 2011.

[17] A. Mittal, A. K. Moorthy, and A. C. Bovik, "No-reference image quality assessment in the spatial domain," *IEEE Trans. Image Process.*, vol. 21, no. 12, pp. 4695–4708, Dec. 2012.

[18] W. Xue, X. Mou, L. Zhang, A. C. Bovik, and X. Feng, "Blind image quality assessment using joint statistics of gradient magnitude and Laplacian features," *IEEE Trans. Image Process.*, vol. 23, no. 11, pp. 4850–4862, Nov. 2014.

[19] K. Gu, G. Zhai, X. Yang, and W. Zhang, "Using free energy principle for blind image quality assessment," *IEEE Trans. Multimedia*, vol. 17, no. 1, pp. 50–63, Jan. 2015.

[20] Q. Li, W. Lin, J. Xu, and Y. Fang, "Blind image quality assessment using statistical structural and luminance features," *IEEE Trans. Multimedia*, vol. 18, no. 12, pp. 2457–2469, Dec. 2016.

[21] K. Gu, D. Tao, J.-F. Qiao, and W. Lin, "Learning a no-reference quality assessment model of enhanced images with big data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 4, pp. 1301–1313, Apr. 2018.

[22] X. Min, G. Zhai, K. Gu, Y. Liu, and X. Yang, "Blind image quality estimation via distortion aggravation," *IEEE Trans. Broadcast.*, vol. 64, no. 2, pp. 508–517, Jun. 2018.

[23] D. Ghadiyaram and A. C. Bovik, "Perceptual quality prediction on authentically distorted images using a bag of features approach," *J. Vis.*, vol. 17, no. 1, p. 32, 2017.

[24] Q. Jiang et al., "Single image super-resolution quality assessment: A real-world dataset, subjective studies, and an objective metric," *IEEE Trans. Image Process.*, vol. 31, pp. 2279–2294, 2022.

[25] L. Kang, P. Ye, Y. Li, and D. Doermann, "Convolutional neural networks for no-reference image quality assessment," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1733–1740.

[26] S. Bosse, D. Maniry, K. Müller, T. Wiegand, and W. Samek, "Deep neural networks for no-reference and full-reference image quality assessment," *IEEE Trans. Image Process.*, vol. 27, no. 1, pp. 206–219, Jan. 2018.

[27] J. Wu, J. Ma, F. Liang, W. Dong, G. Shi, and W. Lin, "End-to-end blind image quality prediction with cascaded deep neural network," *IEEE Trans. Image Process.*, vol. 29, pp. 7414–7426, 2020.

[28] W. Zhang, K. Ma, J. Yan, D. Deng, and Z. Wang, "Blind image quality assessment using a deep bilinear convolutional neural network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 1, pp. 36–47, Jan. 2020.

[29] Z. Pan, F. Yuan, J. Lei, Y. Fang, X. Shao, and S. Kwong, "VCRNet: Visual compensation restoration network for no-reference image quality assessment," *IEEE Trans. Image Process.*, vol. 31, pp. 1613–1627, 2022.

[30] S. A. Golestaneh, S. Dadsetan, and K. M. Kitani, "No-reference image quality assessment via transformers, relative ranking, and selfconsistency," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, Jan. 2022, pp. 1220–1230.

[31] S. Roy, S. Mitra, S. Biswas, and R. Soundararajan, "Test time adaptation for blind image quality assessment," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16696–16705.

[32] Q. Jiang, Z. Liu, S. Wang, F. Shao, and W. Lin, "Toward top-down just noticeable difference estimation of natural images," *IEEE Trans. Image Process.*, vol. 31, pp. 3697–3712, 2022.

[33] F. Li, Y. Zhang, and P. C. Cosman, "MMMNet: An end-to-end multi-task deep convolution neural network with multi-scale and multi-hierarchy fusion for blind image quality assessment," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 12, pp. 4798–4811, Dec. 2021.

[34] S. Yang, Q. Jiang, W. Lin, and Y. Wang, "SGDNet: An end-to-end saliency-guided deep neural network for no-reference image quality assessment," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 1383–1391.

[35] T. Zhou, S. Tan, B. Zhao, and G. Yue, "Multitask deep neural network with knowledge-guided attention for blind image quality assessment," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 8, pp. 7577–7588, Aug. 2024.

[36] K. Bousmalis, G. Trigeorgis, N. Silberman, D. Krishnan, and D. Erhan, "Domain separation networks," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, Aug. 2016, pp. 343–351.

[37] Q. Yang, Y. Liu, S. Chen, Y. Xu, and J. Sun, "No-reference point cloud quality assessment via domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 21147–21156.

[38] B. Sun and K. Saenko, "Deep CORAL: Correlation alignment for deep domain adaptation," in *Proc. Conf. Eur. Conf. Comput. Vis. Workshops (ECCVW)*, Jan. 2016, pp. 443–450.

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.

[39] C. Li, S. Anwar, J. Hou, R. Cong, C. Guo, and W. Ren, "Underwater image enhancement via medium transmission-guided multi-color space embedding," *IEEE Trans. Image Process.*, vol. 30, pp. 4985–5000, 2021.
[40] H. Yan, Y. Ding, P. Li, Q. Wang, Y. Xu, and W. Zuo, "Mind the class weight bias: Weighted maximum mean discrepancy for unsupervised domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 945–954.
[41] H. Li, R. Wan, S. Wang, and A. C. Kot, "Unsupervised domain adaptation in the wild via disentangling representation learning," *Int. J. Comput. Vis.*, vol. 129, no. 2, pp. 267–283, Feb. 2021.
[42] W. Chen, K. Gu, T. Zhao, G. Jiang, and P. L. Callet, "Semi-reference sonar image quality assessment based on task and visual perception," *IEEE Trans. Multimedia*, vol. 23, pp. 1008–1020, 2021.
[43] W. Chen, K. Gu, W. Lin, Z. Xia, P. Le Callet, and E. Cheng, "Reference-free quality assessment of sonar images via contour degradation measurement," *IEEE Trans. Image Process.*, vol. 28, no. 11, pp. 5336–5351, Nov. 2019.
[44] M. Yang and A. Sowmya, "An underwater color image quality evaluation metric," *IEEE Trans. Image Process.*, vol. 24, no. 12, pp. 6062–6071, Dec. 2015.
[45] K. Panetta, C. Gao, and S. Agaian, "Human-visual-system-inspired underwater image quality measures," *IEEE J. Ocean. Eng.*, vol. 41, no. 3, pp. 541–551, Jul. 2016.
[46] Y. Wang et al., "An imaging-inspired no-reference underwater color image quality assessment metric," *Comput. Electr. Eng.*, vol. 70, pp. 904–913, Aug. 2018.
[47] N. Yang, Q. Zhong, K. Li, R. Cong, Y. Zhao, and S. Kwong, "A reference-free underwater image quality assessment metric in frequency domain," *Signal Process., Image Commun.*, vol. 94, May 2021, Art. no. 116218.
[48] Q. Jiang, Y. Gu, C. Li, R. Cong, and F. Shao, "Underwater image enhancement quality evaluation: Benchmark dataset and objective metric," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 9, pp. 5959–5974, Sep. 2022.
[49] Y. Zheng, W. Chen, R. Lin, T. Zhao, and P. Le Callet, "UIF: An objective quality assessment for underwater image enhancement," *IEEE Trans. Image Process.*, vol. 31, pp. 5456–5468, 2022.
[50] M. Yang, Z. Xie, J. Dong, H. Liu, H. Wang, and M. Shen, "Distortionindependent pairwise underwater image perceptual quality comparison," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–15, 2023.
[51] Z. Wang, L. Shen, Z. Wang, Y. Lin, and Y. Jin, "Generation-based joint luminance-chrominance learning for underwater image quality assessment," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 3, pp. 1123–1139, Mar. 2023.
[52] X. Li et al., "Underwater image quality assessment based on multiscale and antagonistic energy," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–14, 2024.
[53] Y. Liu et al., "UIQI: A comprehensive quality evaluation index for underwater images," *IEEE Trans. Multimedia*, vol. 26, pp. 2560–2573, 2024.
[54] W. Hou, X. Gao, D. Tao, and X. Li, "Blind image quality assessment via deep learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 26, no. 6, pp. 1275–1286, Jun. 2015.
[55] X. Liu, J. Van De Weijer, and A. D. Bagdanov, "RankIQA: Learning from rankings for no-reference image quality assessment," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1040–1049.
[56] Y. LeCun et al., "Handwritten digit recognition with a back-propagation network," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*. San Mateo, CA, USA: Morgan Kaufmann, 1990, pp. 396–404.
[57] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.
[58] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
[59] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[60] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Med. Image Comput. Comput.-Assist. Intervent (MICCAI)*, Jan. 2015, pp. 234–241.
[61] S. Guo, Z. Yan, K. Zhang, W. Zuo, and L. Zhang, "Toward convolutional blind denoising of real photographs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1712–1722.
[62] W. Ren et al., "Low-light image enhancement via a deep hybrid network," *IEEE Trans. Image Process.*, vol. 28, pp. 4364–4375, 2019.
[63] D. Berman, D. Levy, S. Avidan, and T. Treibitz, "Underwater single image color restoration using haze-lines and a new quantitative dataset," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 8, pp. 2822–2837, Aug. 2021.
[64] A. Galdran, D. Pardo, A. Picón, and A. Alvarez-Gila, "Automatic red-channel underwater image restoration," *J. Vis. Commun. Image Represent.*, vol. 26, pp. 132–145, Jan. 2015.
[65] C. Li et al., "An underwater image enhancement benchmark dataset and beyond," *IEEE Trans. Image Process.*, vol. 29, pp. 4376–4389, 2020.
[66] J.-S. Lee, "On designing paired comparison experiments for subjective multimedia quality assessment," *IEEE Trans. Multimedia*, vol. 16, no. 2, pp. 564–571, Feb. 2014.
[67] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
[68] W. Zhang, Y. Liu, C. Dong, and Y. Qiao, "RankSRGAN: Generative adversarial networks with ranker for image super-resolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3096–3105.
[69] Y. Guo, H. Li, and P. Zhuang, "Underwater image enhancement using a multiscale dense generative adversarial network," *IEEE J. Ocean. Eng.*, vol. 45, no. 3, pp. 862–870, Jul. 2020.

Qiuping Jiang (Senior Member, IEEE) is currently a Professor at the School of Information Science and Engineering, Ningbo University, China. His research interests include image quality assessment, visual perception modeling, and water-related vision. He serves as an Associate Editor for *Pattern Recognition*, *Displays*, *Journal of Visual Communication and Image Representation*, and *Journal of Electronic Imaging*, and the Area Chair for IEEE ICME (2022–2025) and IEEE ICIP (2025).

Yuese Gu received the master's degree from the School of Information Science and Engineering, Ningbo University, Ningbo, China, in June 2023. His research interests include image quality assessment and water-related vision.

Zongwei Wu (Member, IEEE) received the French engineering degree (master's degree) in mechanical engineering from the University of Technologies of Compiègne-Sorbone University Alliance in 2019 and the Ph.D. degree in computer vision from the University of Burgundy in 2022. He is currently a Postdoctoral Researcher with the University of Würzburg, Germany. His research interests include multimodal models and image processing.

Chongyi Li (Senior Member, IEEE) was a Research Assistant Professor at Nanyang Technological University, Singapore, and a Research Fellow at Nanyang Technological University and the City University of Hong Kong. He is currently a Professor at Nankai University, Tianjin, China. His research interests include low-level vision and computational photography.

Huan Xiong received the B.S. and M.S. degrees from the School of Mathematical Sciences, Peking University, China, in 2010 and 2013, respectively, and the Ph.D. degree from the Institute of Mathematics, University of Zurich, Switzerland, in 2016. He is currently a Full Professor with Harbin Institute of Technology, China, and an Assistant Professor with the Mohamed bin Zayed University of Artificial Intelligence, Abu Dhabi, United Arab Emirates.

Feng Shao (Member, IEEE) received the B.E. and Ph.D. degrees from Zhejiang University, China, in 2002 and 2007, respectively. He is currently a Professor with the School of Information Science and Engineering, Ningbo University, China. He has published over 100 technical papers in refereed journals and conferences in the areas of image processing, image quality assessment, and immersive media computing.

Zhihua Wang (Member, IEEE) received the B.E. degree from China University of Mining and Technology, Xuzhou, China, in 2014, the M.S. degree from the National University of Defense Technology, Changsha, China, in 2016, and the Ph.D. degree from the Department of Computer Science, City University of Hong Kong, Hong Kong, in 2022. His research interests include image quality assessment and computational vision.

Authorized licensed use limited to: Shiv Nadar University Chennai. Downloaded on September 21,2026 at 05:24:46 UTC from IEEE Xplore. Restrictions apply.